

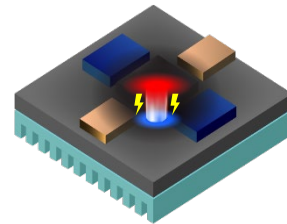
Electrical Power Generation from Microsystem Waste Heat

Reclaiming all tappable energy with embedded thermal corralling

DARPA-SN-25-69

Addendum

April 2025





Outline

1. Motivation for interest
2. Limiting factors
 - a) Reclamation efficiency potential drops steeply due to heat spreading
 - b) Current reclamation devices cannot capture heat where it is generated in microsystems
3. Developing technologies
 - a) Required advancements and high-level approaches to address limiting factors
 - b) Additional technologies possibly poised to address further concerns within microsystems
4. Performance potential from advancements in energy reclamation
5. Potential impact from microsystem energy harvesting



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We are leaving the greatest readily available source of electrical power on the table

Tapped

Reclaimable electricity from US waste heat^[1]

Untapped Microsystem scavenging

Boilers for power generation from a cement kiln^[2]



~1300 °C

>650 °C,
High-grade,
25 TWh

230-650 °C,
Medium-grade,
63 TWh

<230 °C,
Low-grade,
84 TWh

**Enough
to power
Maryland**

Chip

Substrate

Heat sink

Microsystems as low-grade heat sources:

- Central Processing Units (CPUs)
- Graphics Processing Unit (GPUs)
- Radio frequency (RF) amplifiers

Thermoelectric generators in automotive exhaust system^[3]

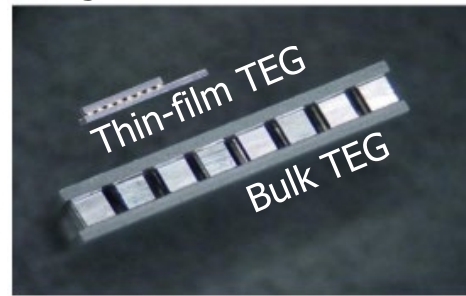
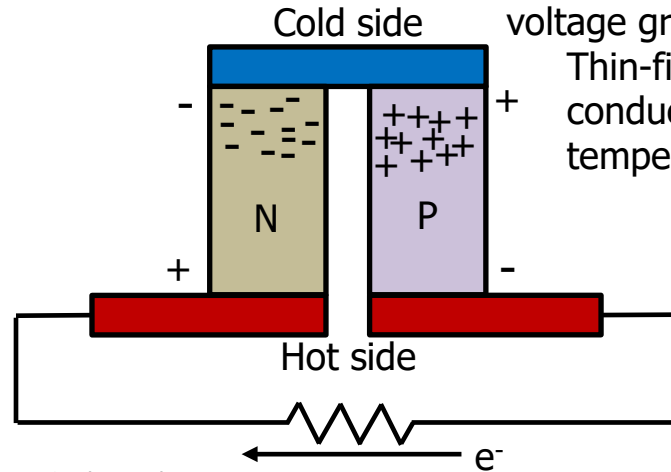


~300 °C

Low-grade waste heat provides great potential anywhere microsystems are present

Thin-film thermoelectrics ($V \sim 0.01 \text{ cm}^2$)

Directly converts a temperature gradient to a voltage gradient through the Seebeck effect
Thin-film TEGs have a lower thermal conductivity, enabling a greater temperature gradient

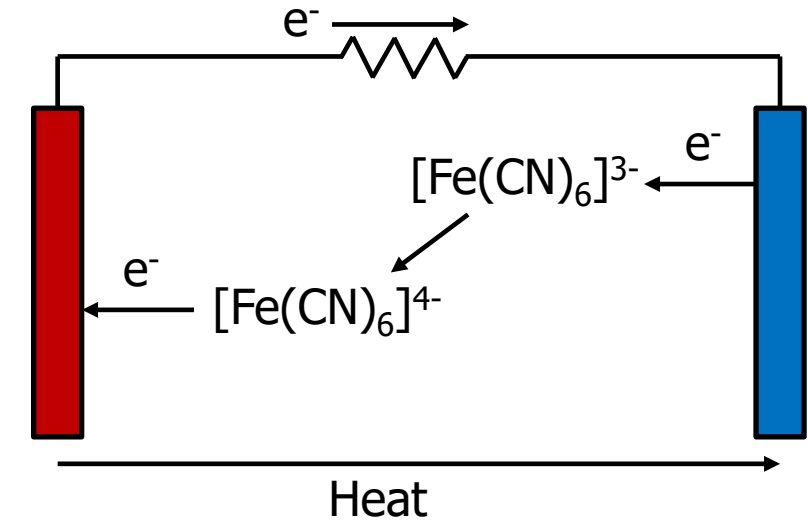


Laird Technologies, *Thin Film Thermoelectric Handbook*.

Thermogalvanic cell ($V \sim 10 \text{ cm}^2$)

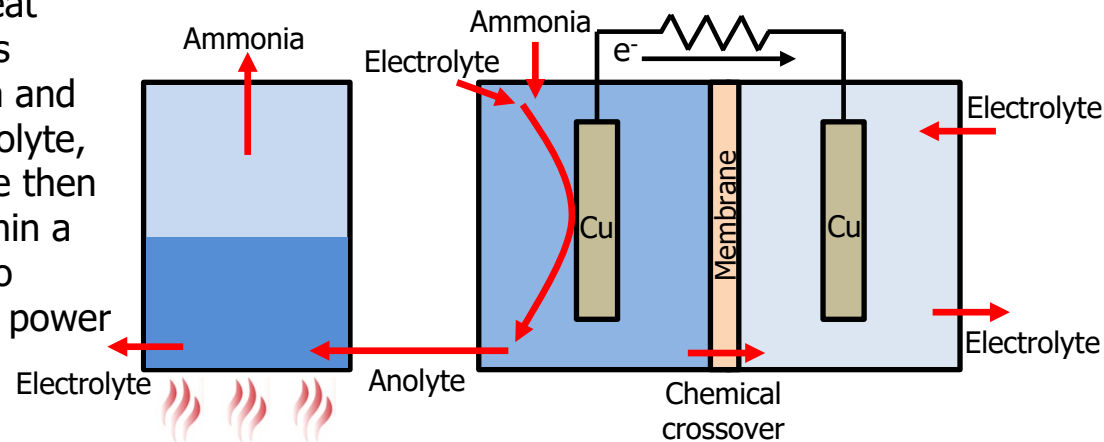
A temperature gradient causes a chemical equilibrium gradient

Electrons flow along the path of least resistance towards equilibrium



Thermally regenerative battery ($V \sim 1 \text{ cm}^2$)

Waste heat separates ammonia and an electrolyte, which are then used within a battery to generate power

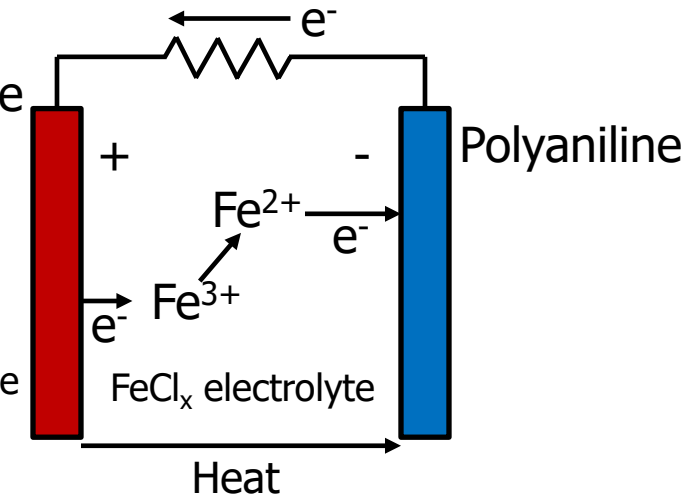


Direct thermal conversion cell ($V \sim 10 \text{ cm}^2$)

Pt/Graphene Oxide

Same operating principle as the thermogalvanic cell above

Optimized with a "temperature-induced pseudocapacitive effect" provided by the graphene oxide, which increases the voltage achieved





Prior device level heat removal/reclamation investigations have not addressed all needs for efficient reclamation

	ICECool	MATRIX	THREADS	Minitherms3D	Special notice
Goal	Embedded, microfluidic cooling	Integrate advanced transduction materials	Realize transistors that operate near the limit of RF output power	Enabling 3DHI with internal and external thermal management	Efficient conversion of waste heat to useful electricity
Embedded thermal corralling	Partial	No	Yes	Partial	Yes
Thermal path type	Spreading	N/A	Spreading	Spreading	Directed
Heat reclamation	No	Partial	No	No	Yes
Heat source temperature	N/A	>300°C	N/A	N/A	<230°C
Lowest integration level	Chip	System	Chip	Chip	Chip



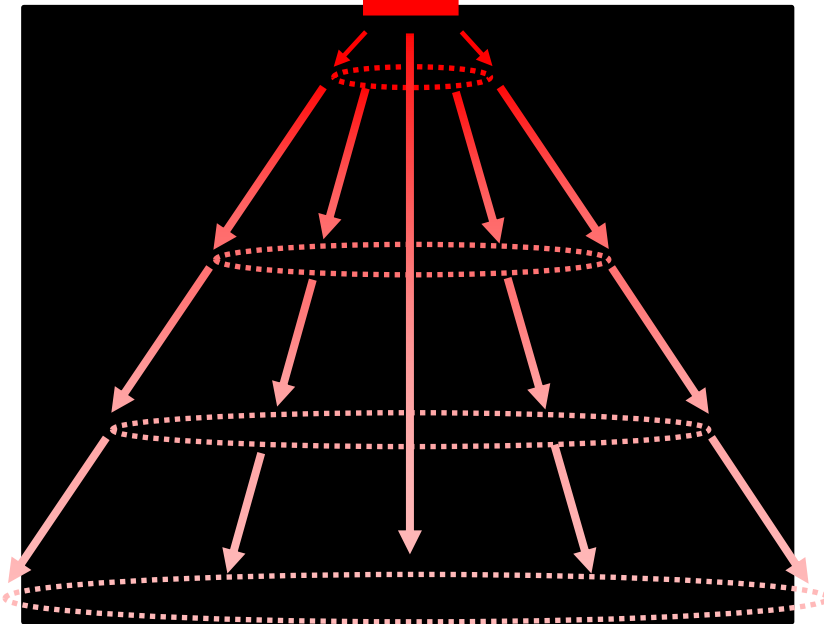
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Reclamation limited by unabated heat spreading and large devices

Uncorralled, heat spreads unabated

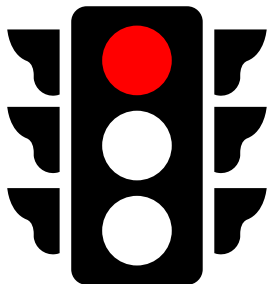
Hotspot within a chip



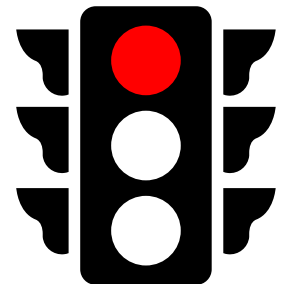
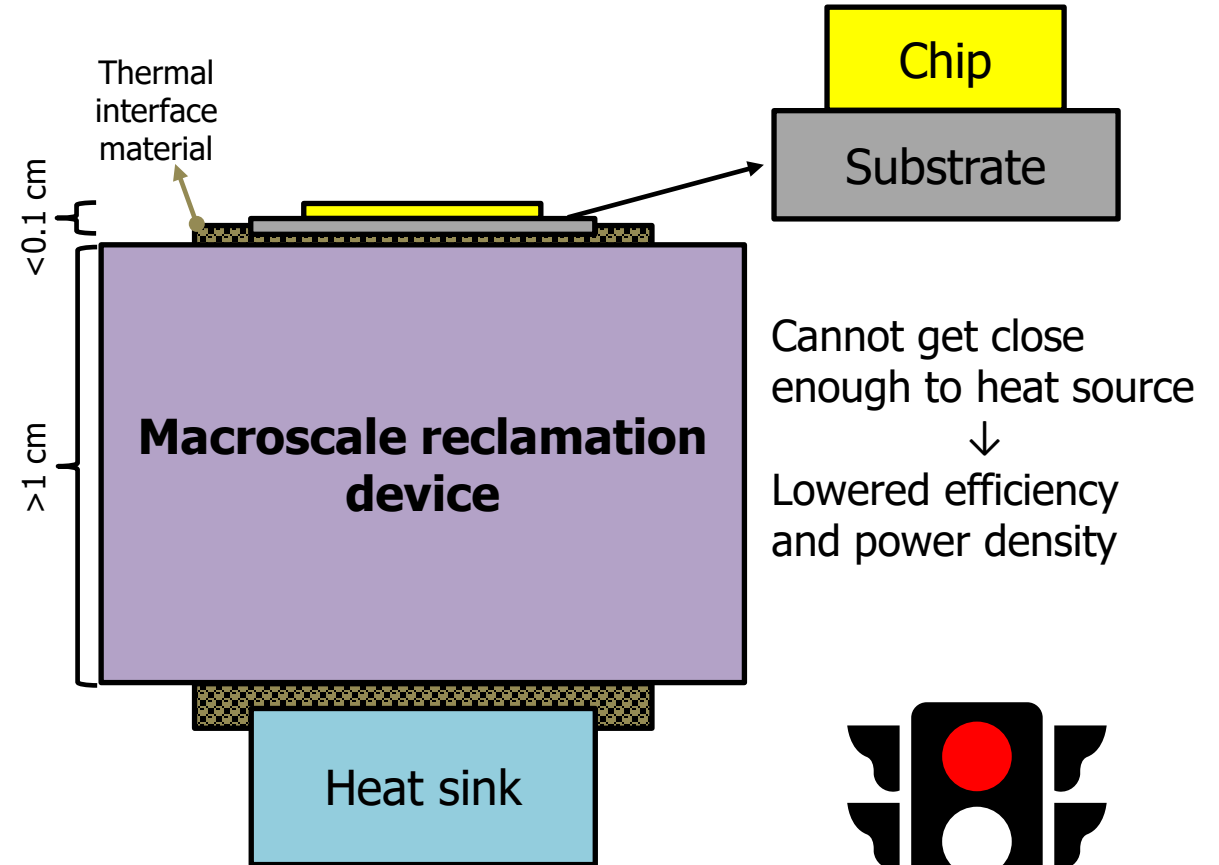
Rapid temperature reduction
from top to bottom



Reduced reclamation efficiency



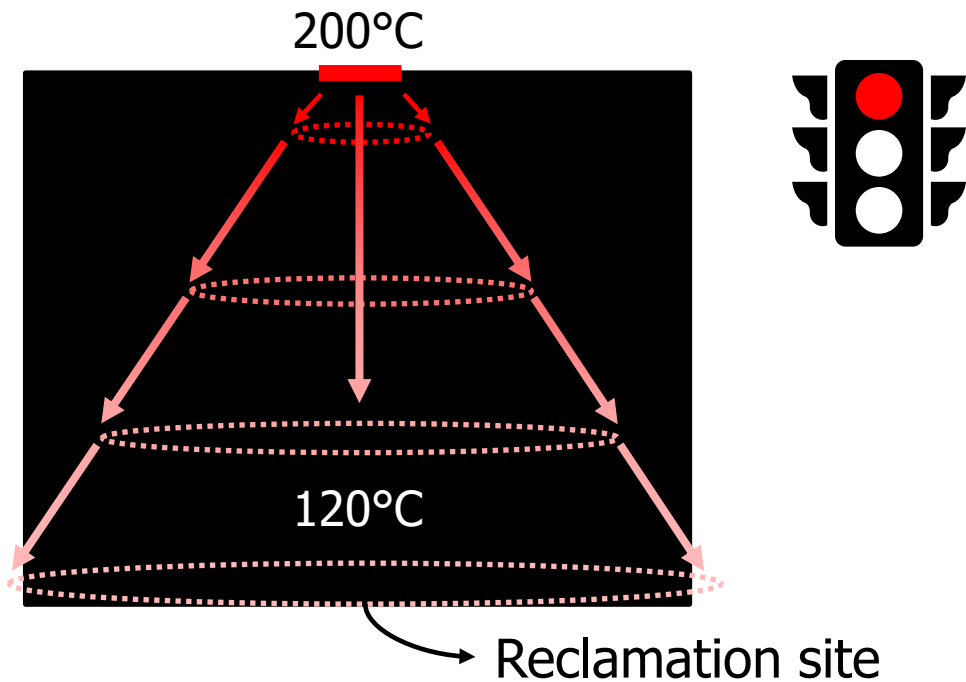
Current reclamation is too large





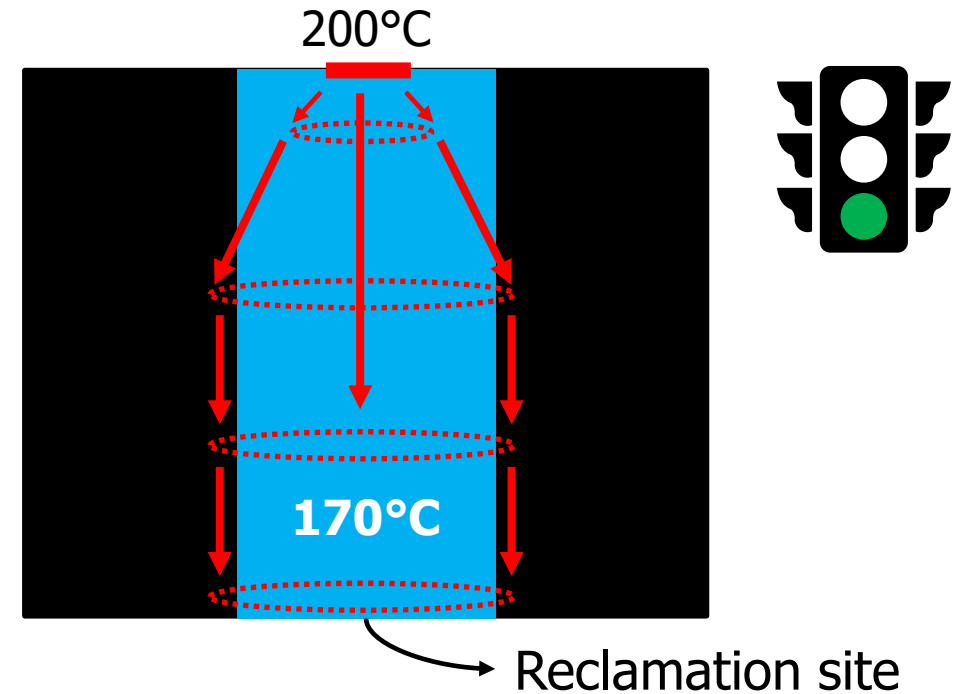
Heat corralling has potential to improve efficiency to near the thermodynamic limit

Low-grade heat spreading (before)



Achieved relative efficiency ~25%

Low-grade heat focusing (after)



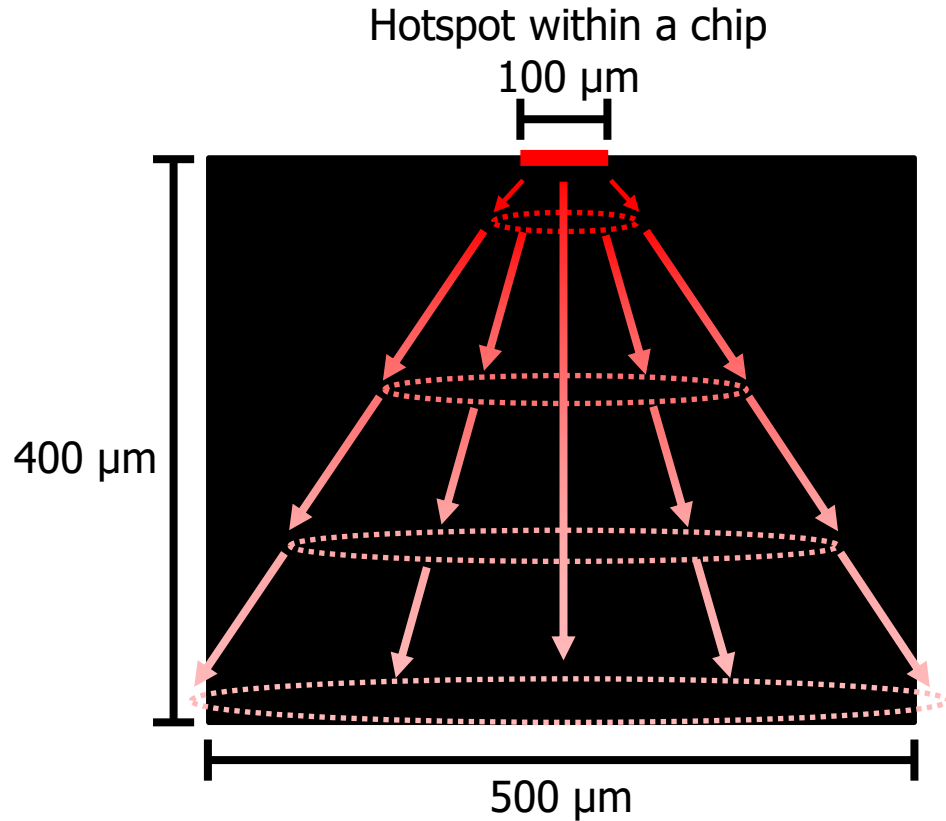
Potential relative efficiency ~80%

Si substrate with thermal conductivity of 140 W/mK

Si substrate with thermal conductivity of 140 W/mK
Via conductivity of 1000 W/mK and diameter of 200 μ m

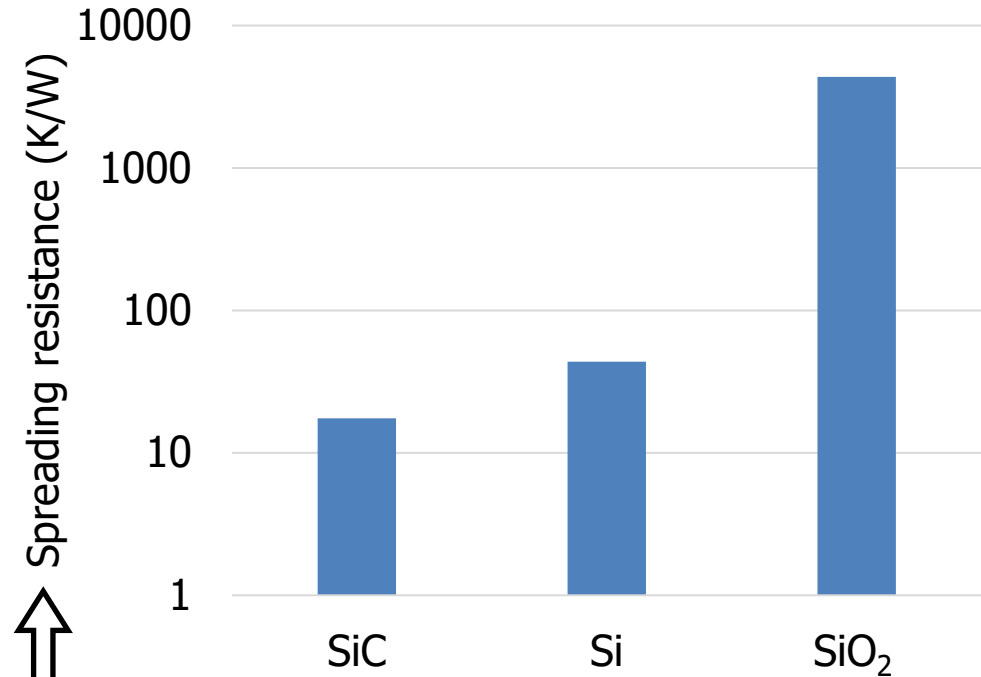


How hard is low-grade heat corralling?



$$R_{spreading} = \frac{H}{\pi k r_{hotspot}}$$

- H is a function of the geometry
- k is the thermal conductivity of the material
- $r_{hotspot}$ is the radius of the hotspot (50 μm)



- SiC and Si have relatively **high thermal conductivity** (>100 W/mK) that is nearly isotropic (same in all directions), so they **rapidly spread heat, lowering efficiency**
- Insulators like SiO₂ have **low thermal conductivity** and **do not diffuse heat rapidly**



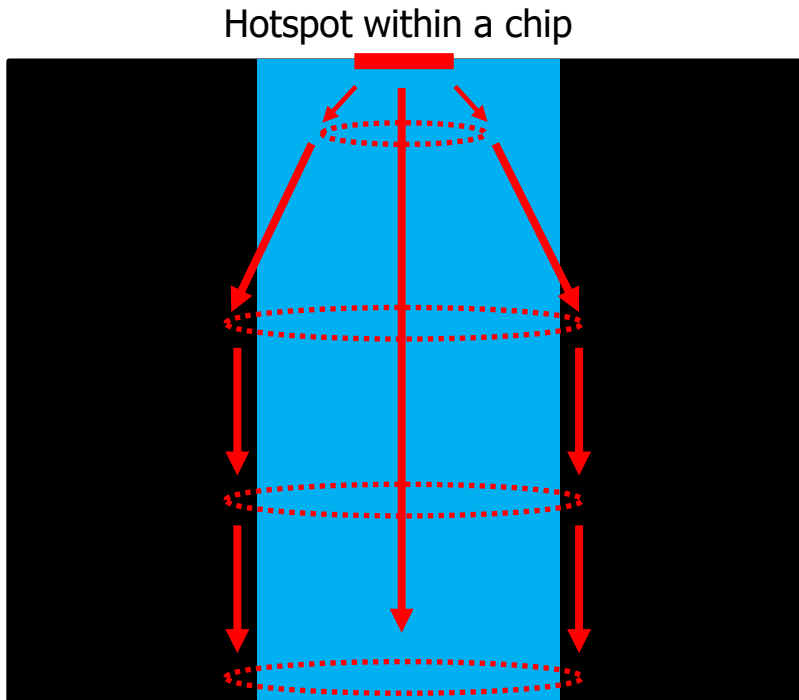
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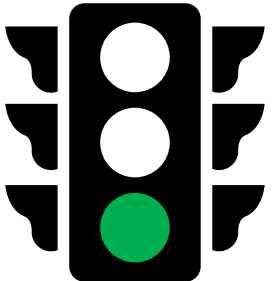


Even with corralled heat, need devices that can reach it are required

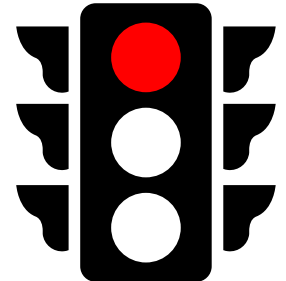
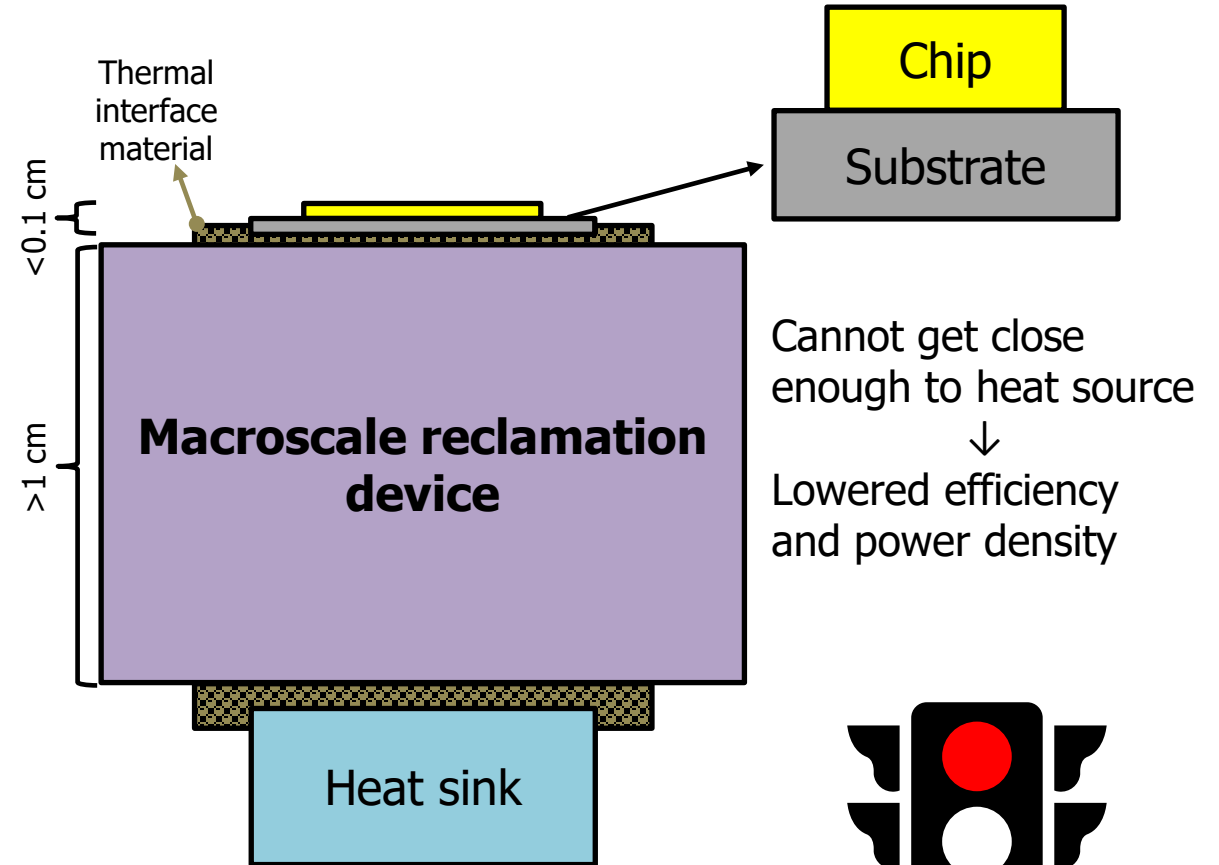
Corralled heat remains focused



Small temperature drop
maintained from top to bottom
↓
Minimal reclamation potential loss



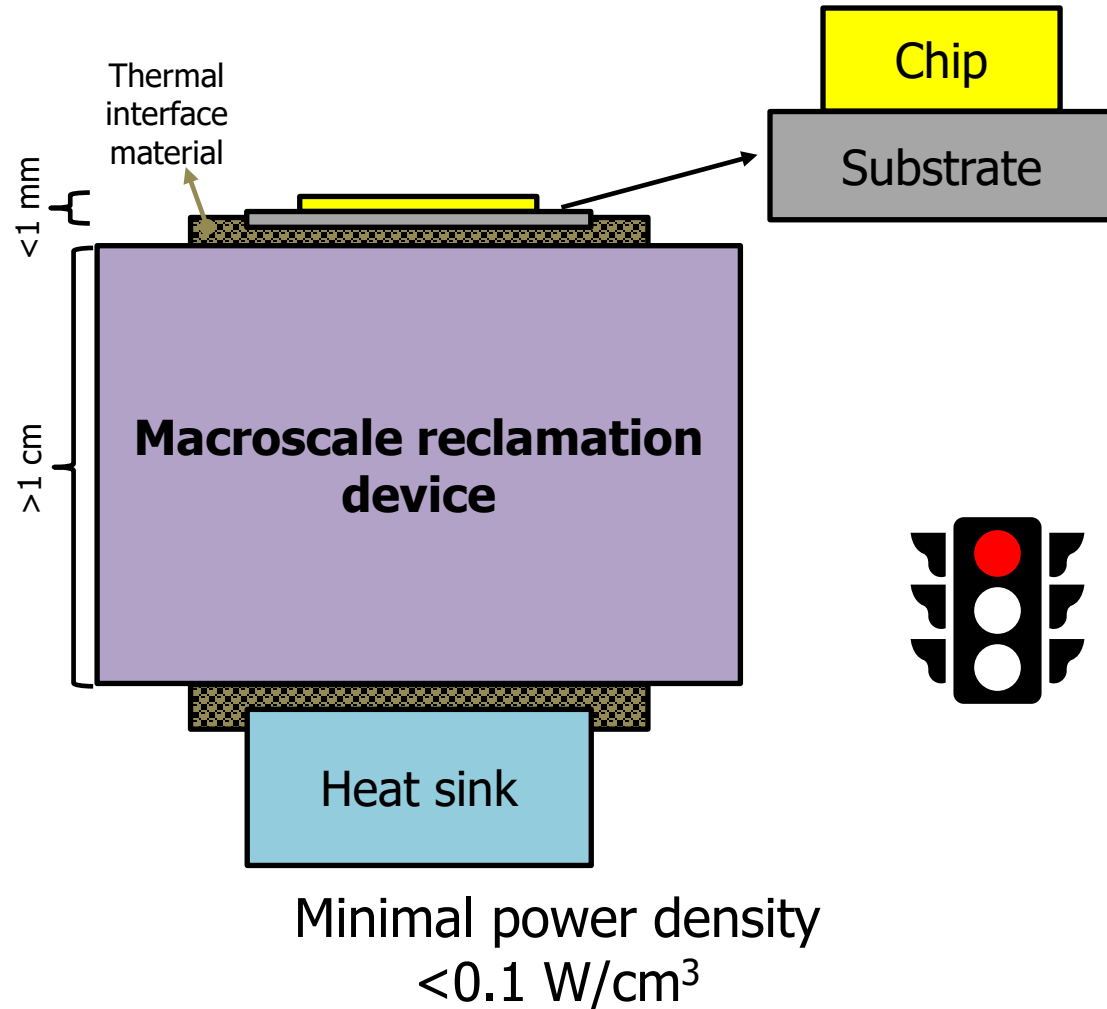
Current reclamation is too large



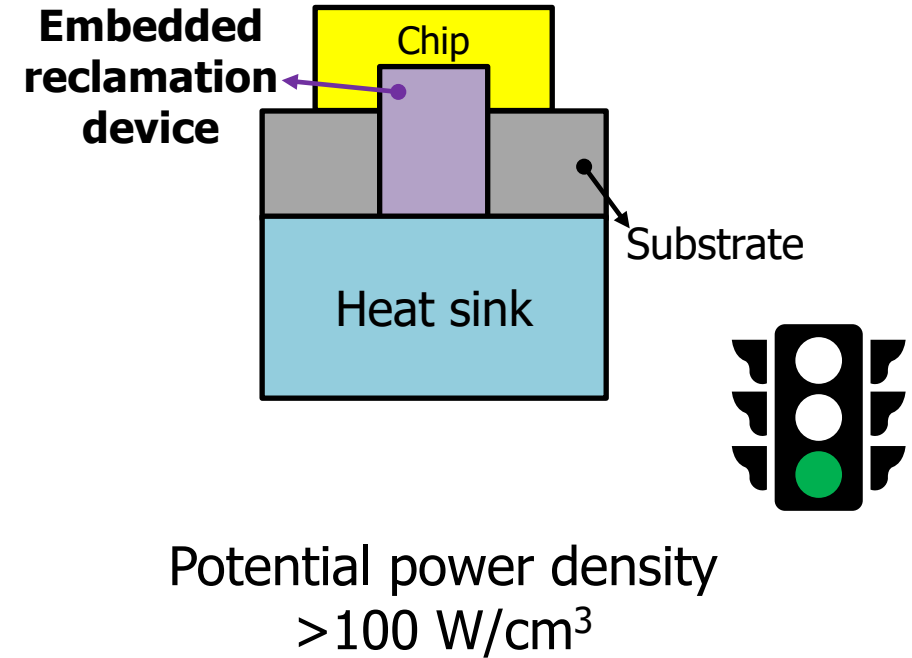


Embedded reclamation achieves highest efficiency and power density

Macroscale reclamation

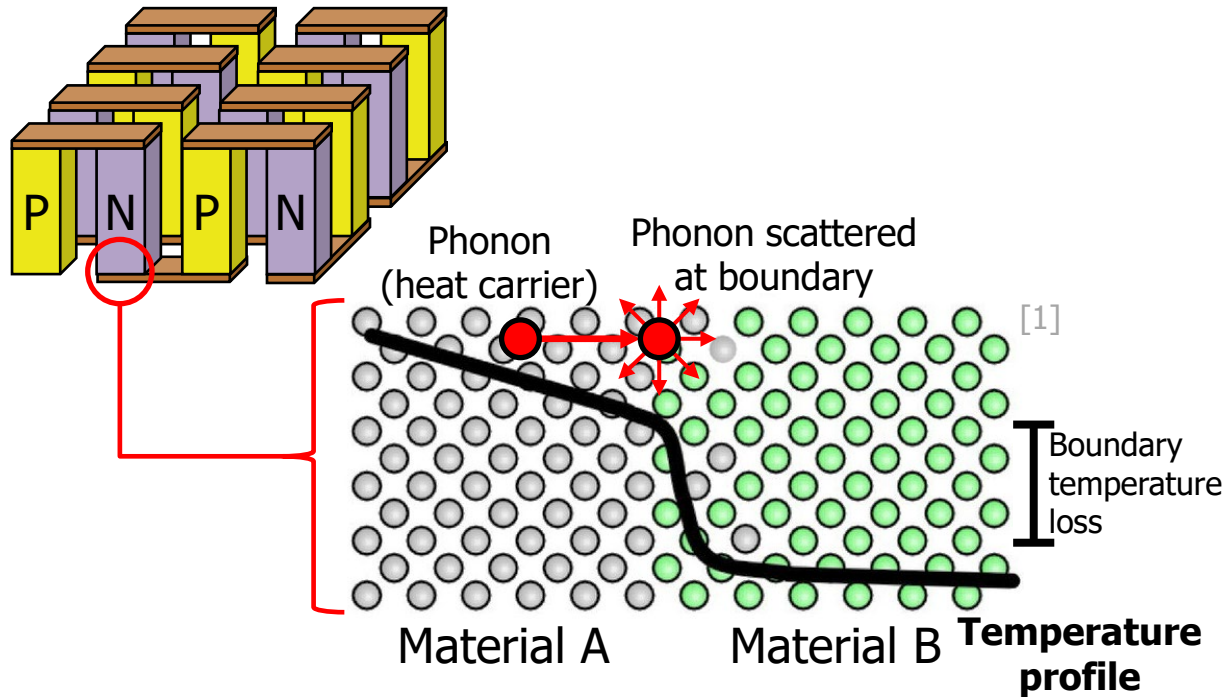


Embedded reclamation



How hard is embedded reclamation?

SOLID-STATE

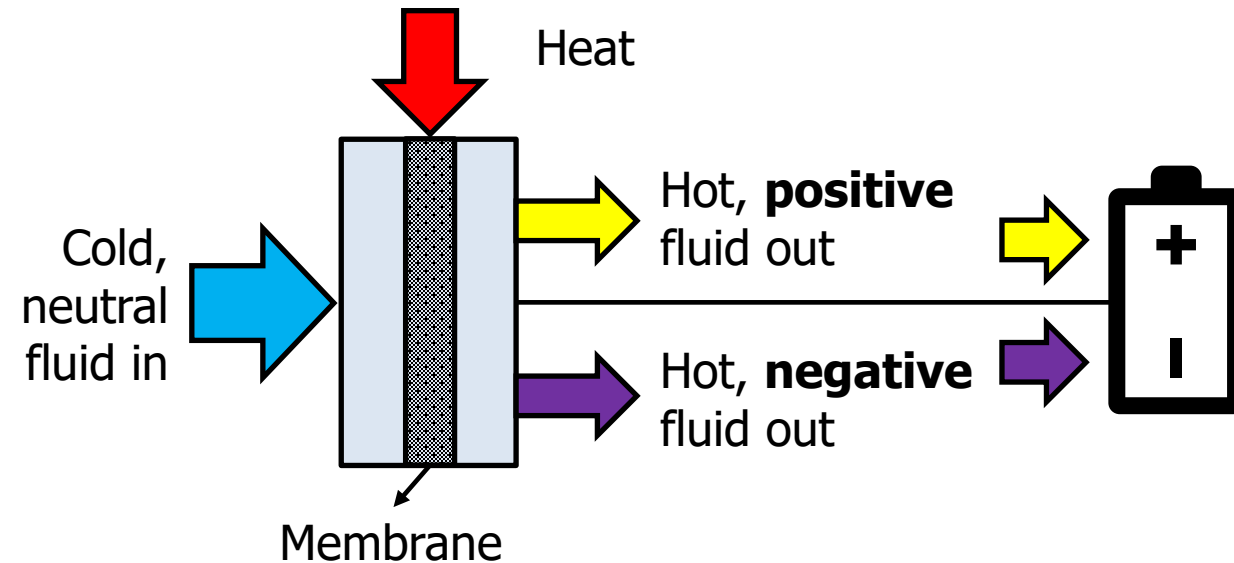


$$Q = G\Delta T$$

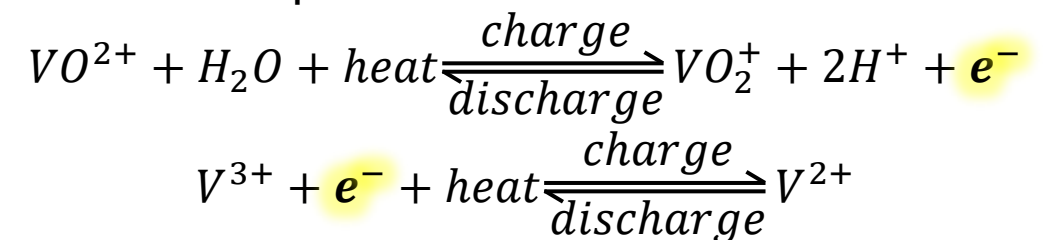
- Q is the heat flow through the boundary
- G is the thermal boundary conductance (experimentally or numerically derived)
- ΔT is the boundary temperature loss

CHEMICAL

Must balance complex, multi-stream microfluidics with dynamic chemical reactions



Example chemical reaction set^[2]:





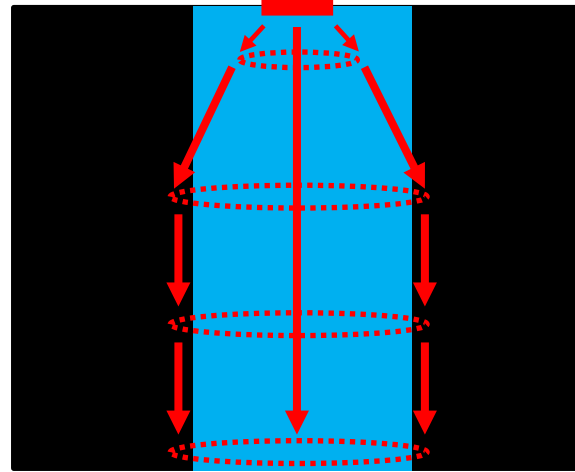
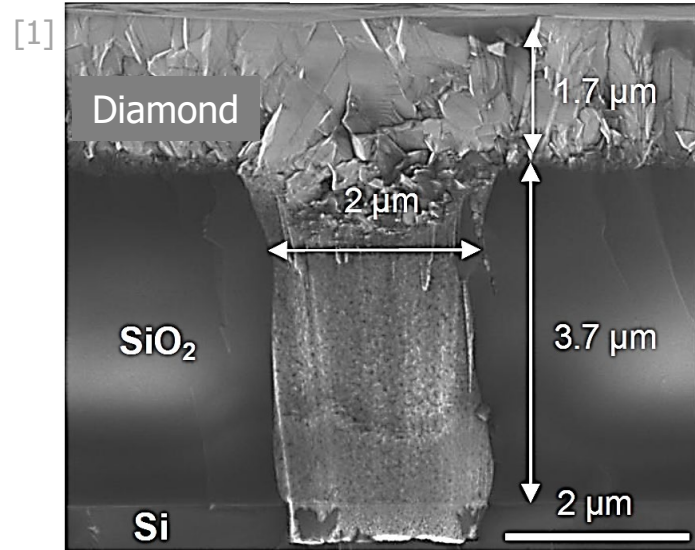
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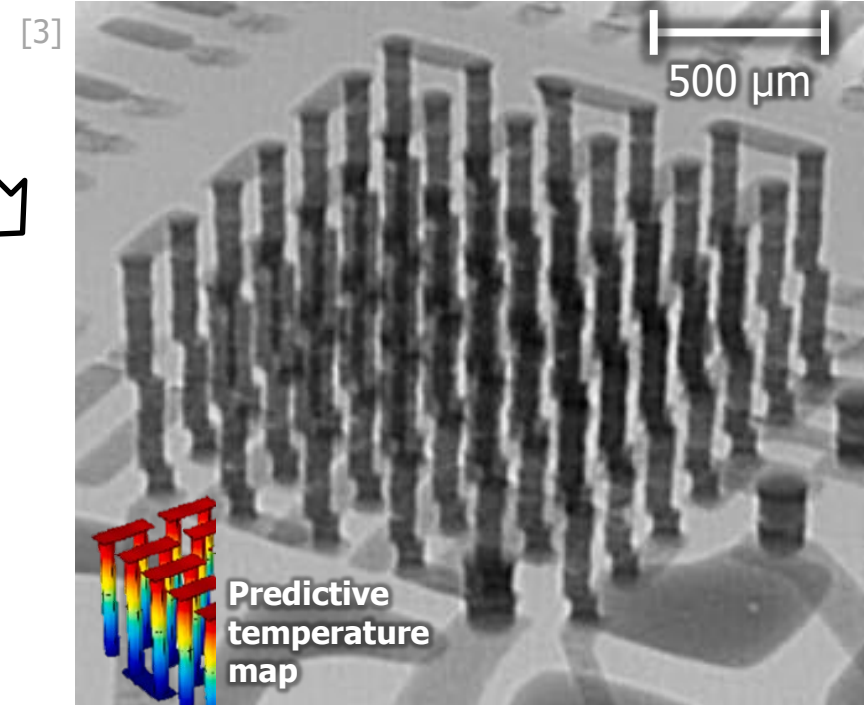


Heat corralling requires material and design advancement

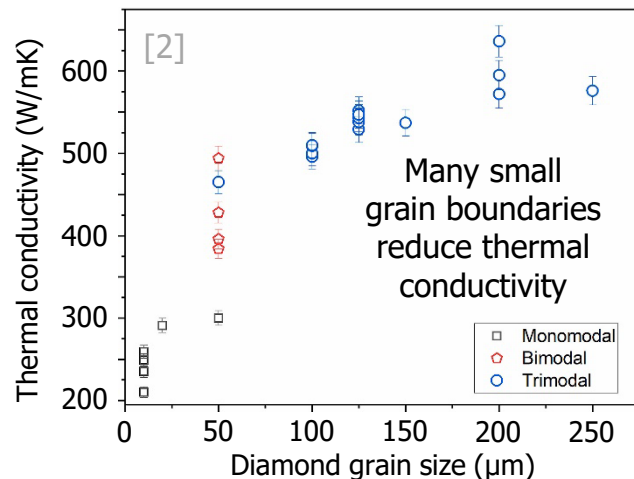
MATERIAL



DESIGN



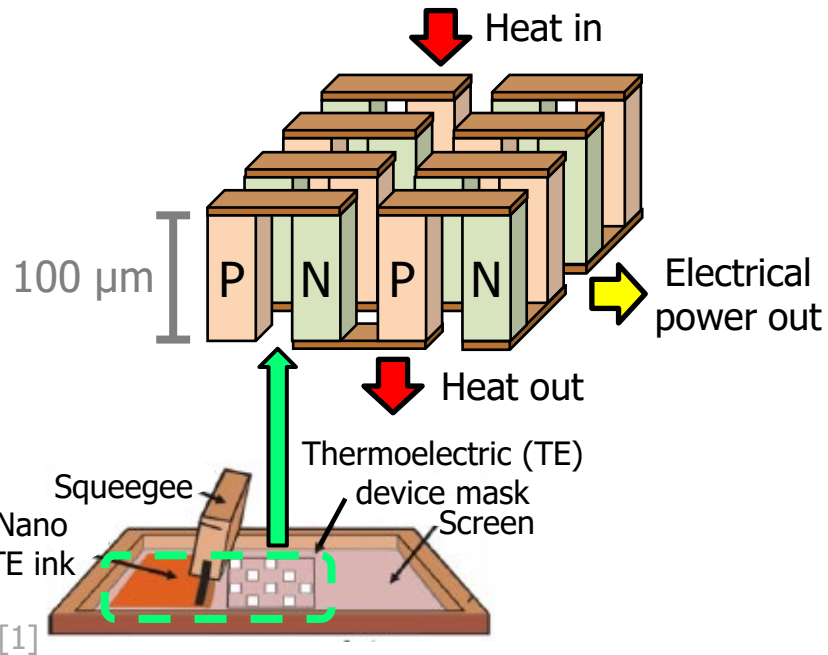
Challenge: Must integrate corralling structures with back-end power delivery without degrading chip performance



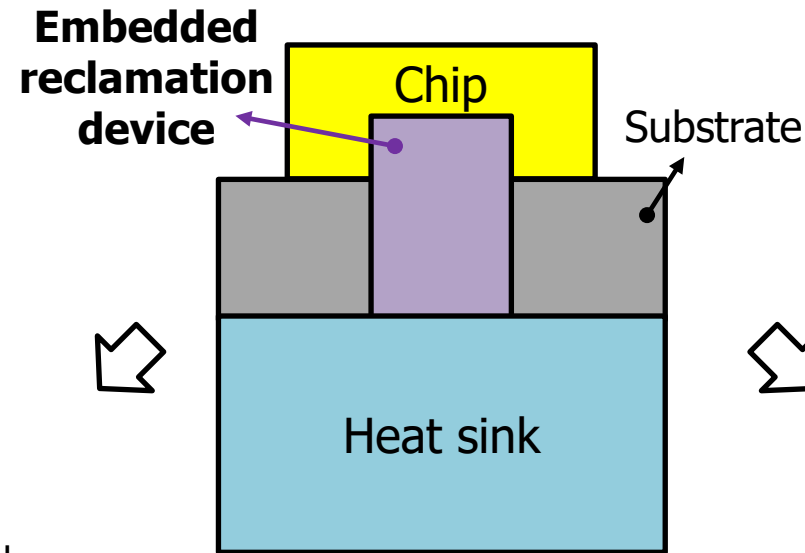
Challenge: Need to control thermal properties in lateral and vertical directions for efficient corralling with electrically insulative materials

SOLID-STATE

On-wafer printing of solid-state reclamation

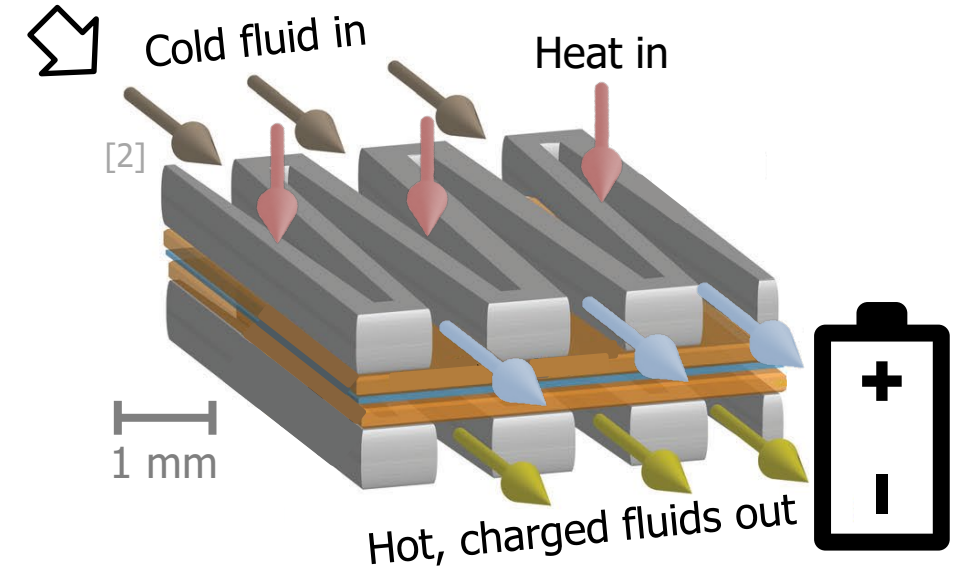


Challenge: Maintain low thermal resistance at all material interfaces for high conversion efficiency



CHEMICAL

Fluidic systems for chemical reclamation



Challenge: Requires complex interaction of fluid mechanics and chemical reaction dynamics within a dense package



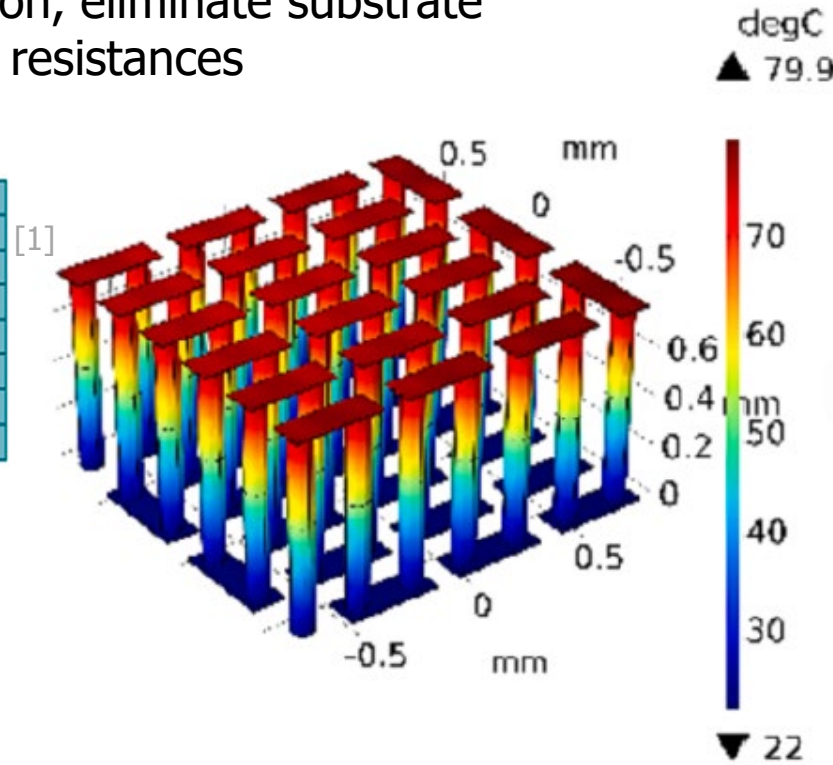
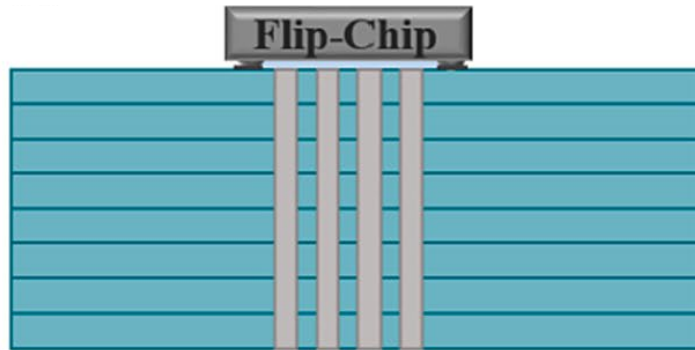
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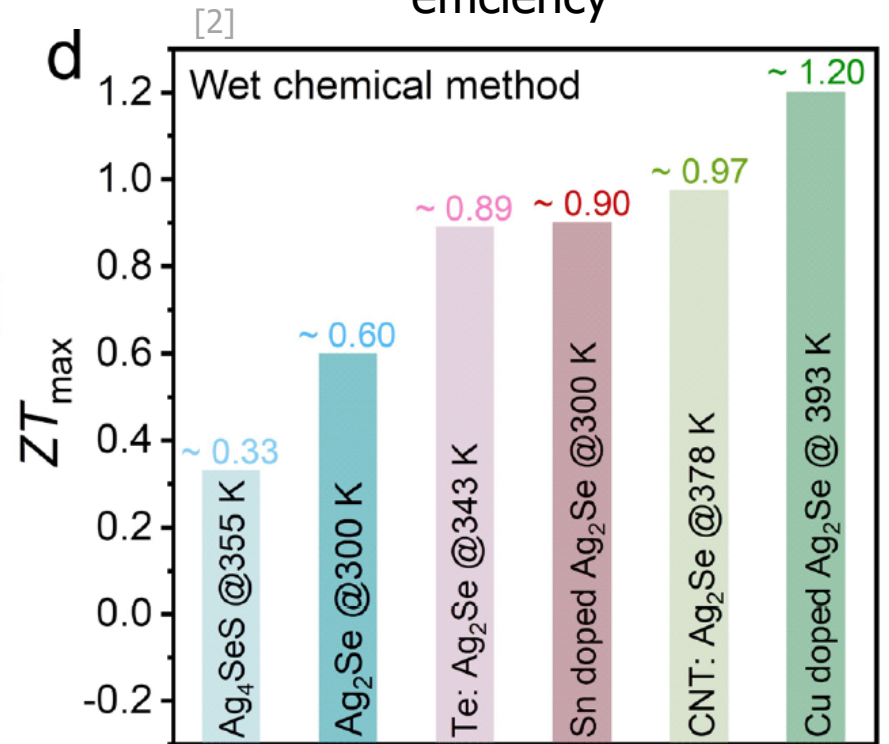


Embedded reclamation introduces new thermal boundaries/interfaces

Ag-based thermocouple vias, for simultaneous capture and conversion, eliminate substrate boundary resistances



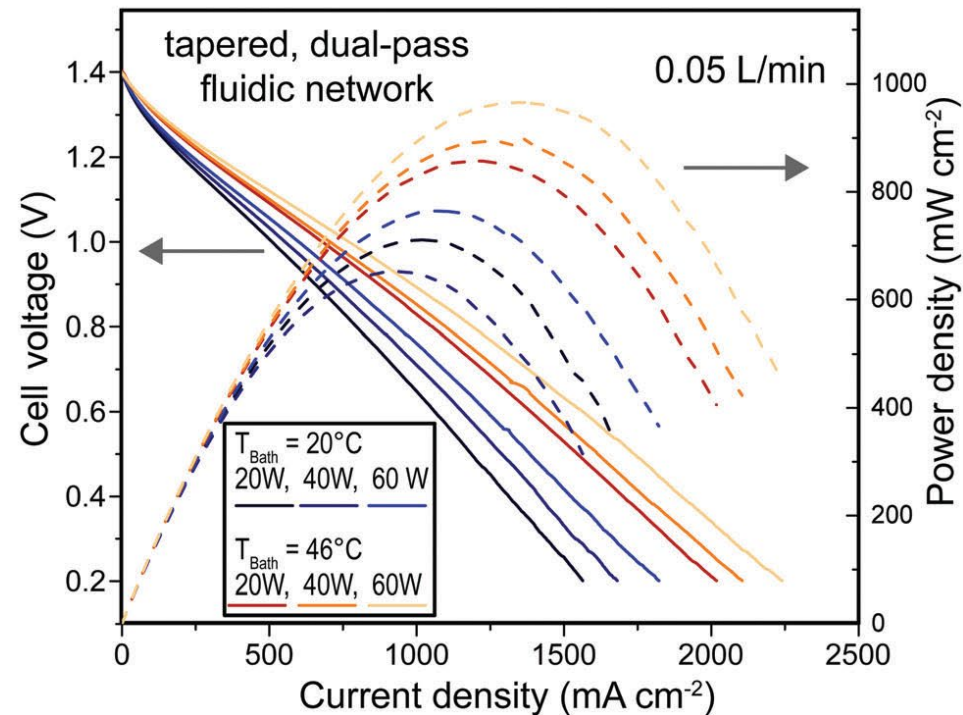
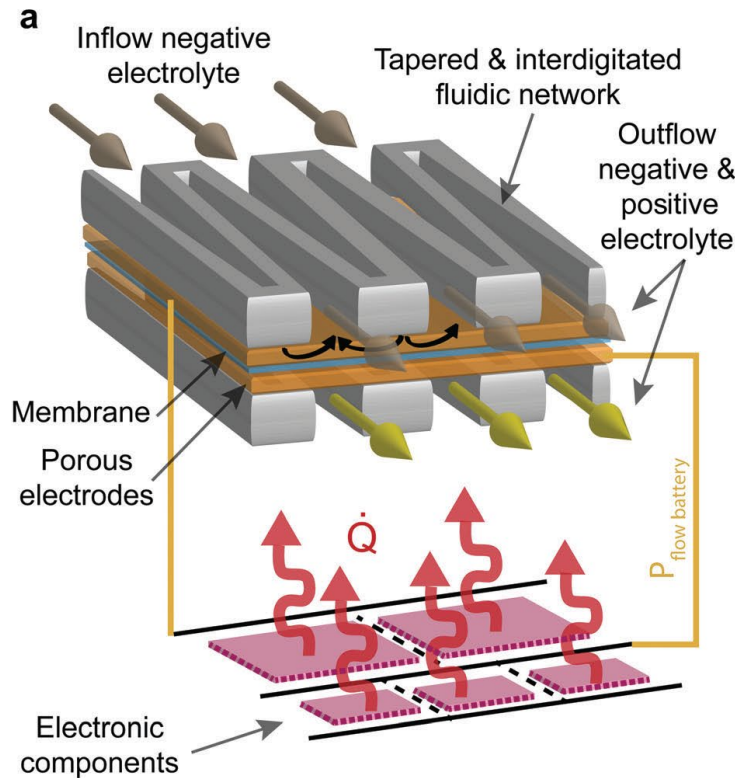
Poor conversion efficiency



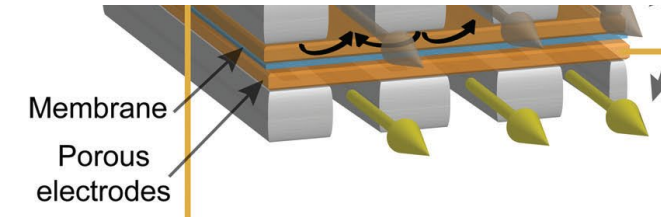
[1] N. Jaziri et al., *Energy Conversion and Management*, 2024

[2] H. Wu et al., *Energy & Environmental Science*, 2023

3D-printed redox flow battery for embedded electronics cooling



Membrane is a major resistance, limiting performance

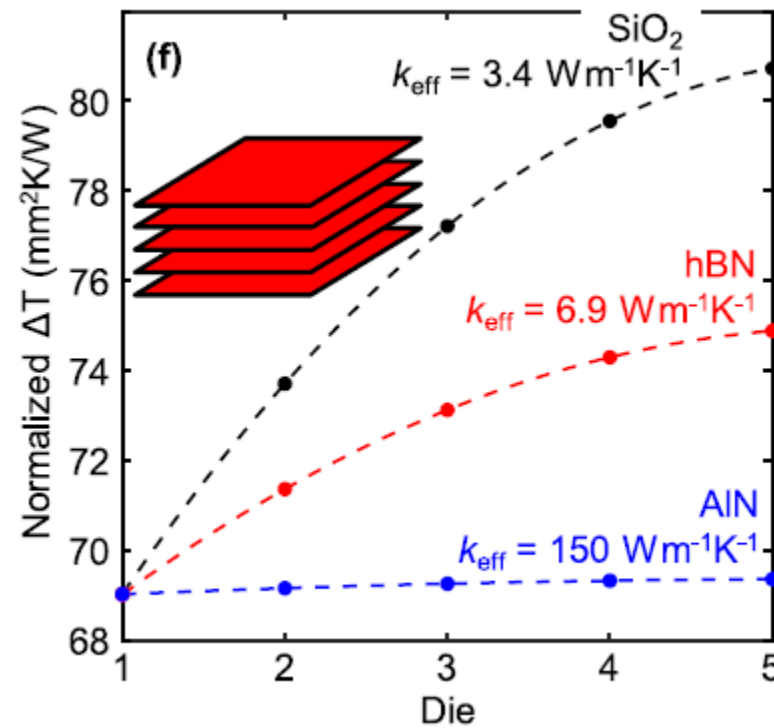
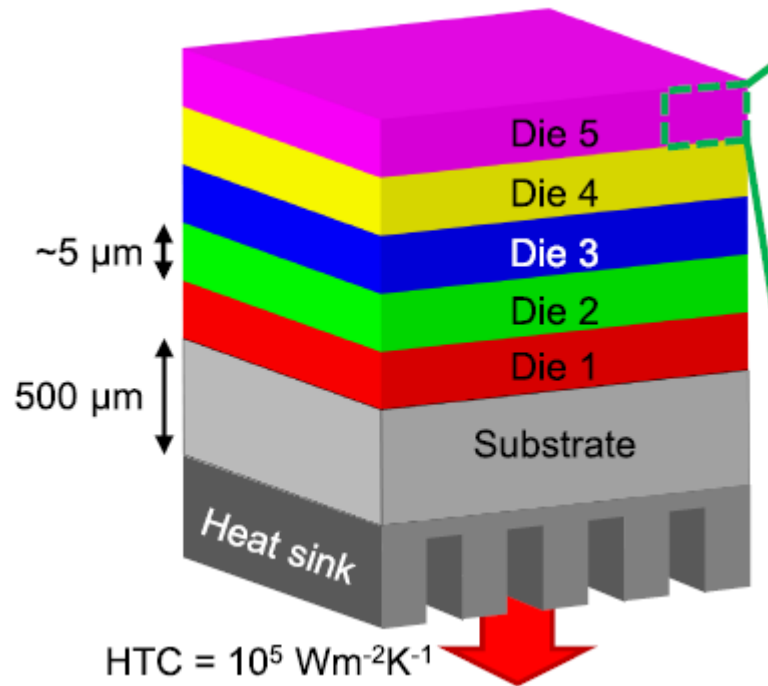


3D-printing technologies are still under development and are slow in fabrication speed



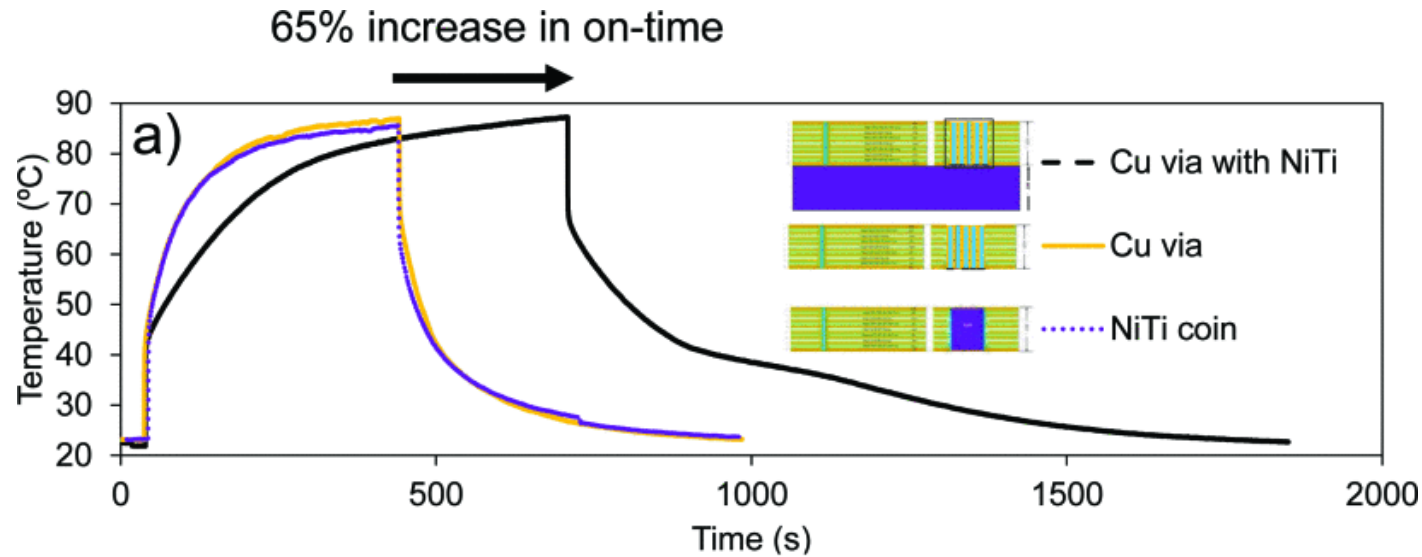
Reclamation device adds thermal resistance

New dielectrics with high thermal conductivity (AlN and hexagonal BN)

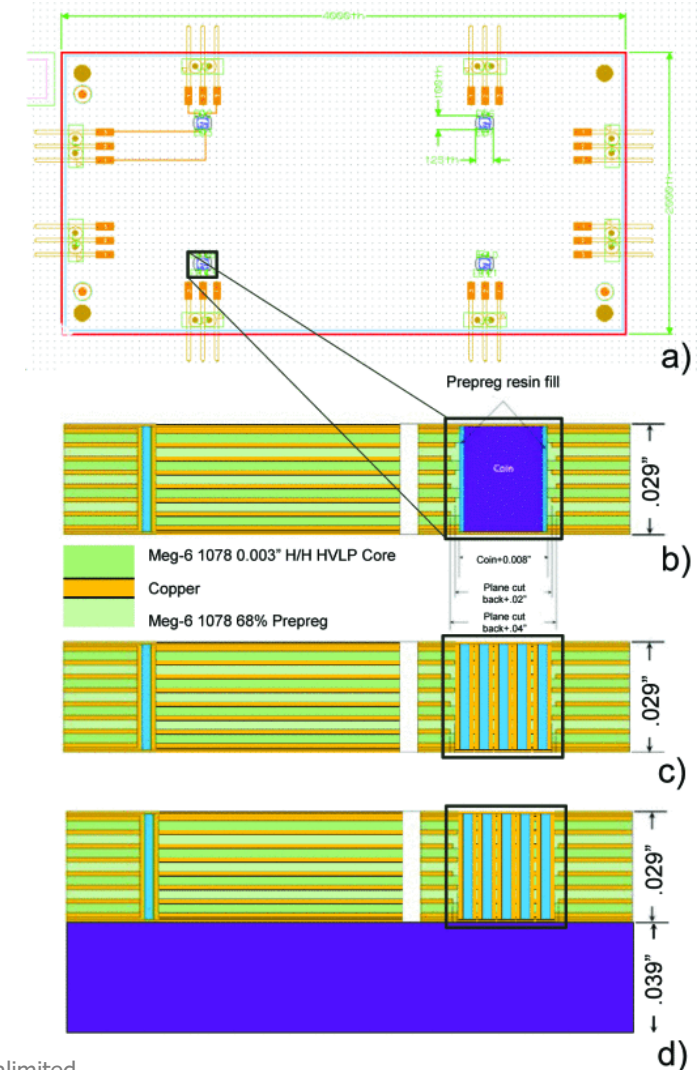


Fabrication processes and experimental validation incomplete

Embedded solid-solid PCMs for dynamic capture



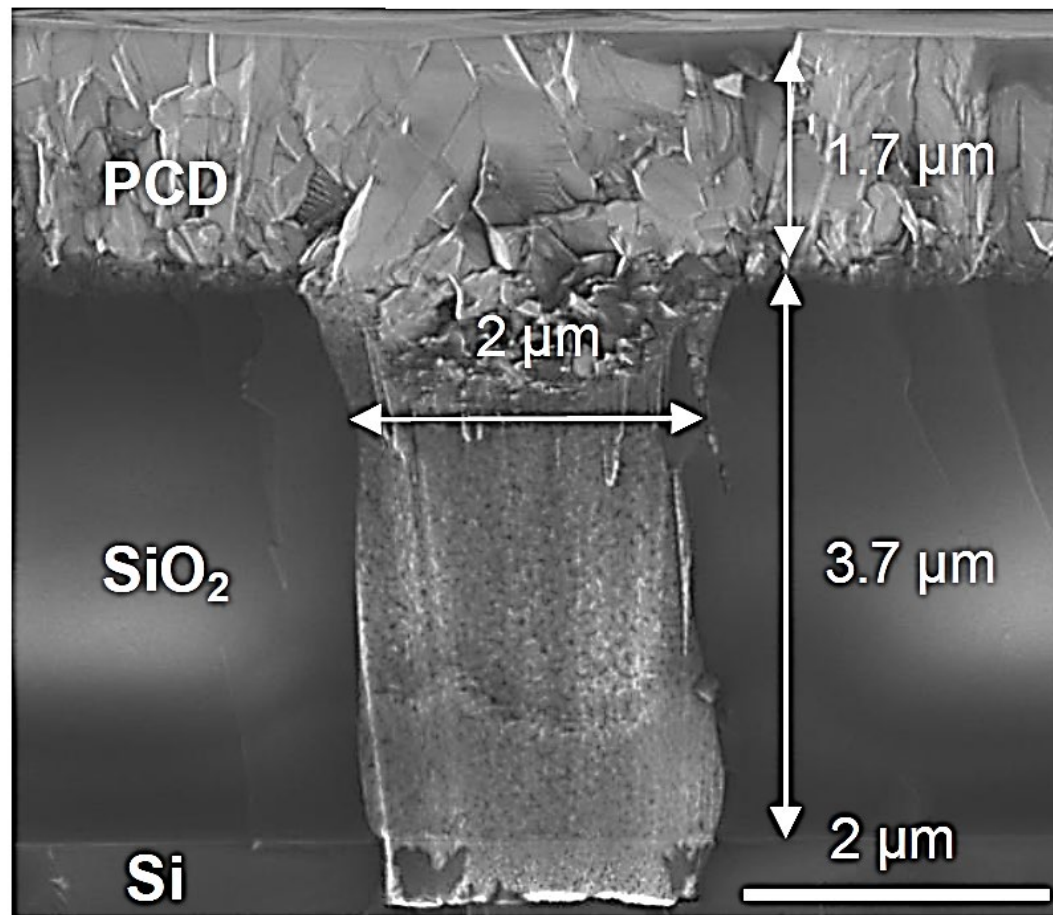
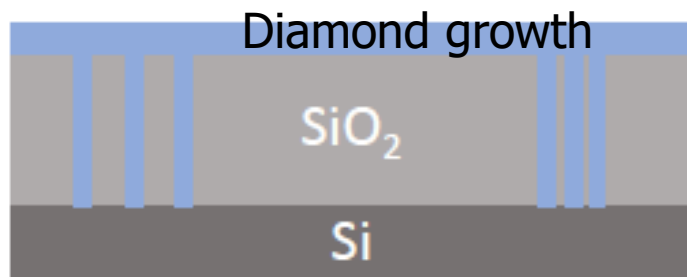
PCM must be as close to the junction as possible to maximize dynamic capture





Microsystem heat diffuses across the substrate prior to capture

Diamond thermal via growth for directed (non-spreading) thermal capture



Low aspect ratio
(~2:1)

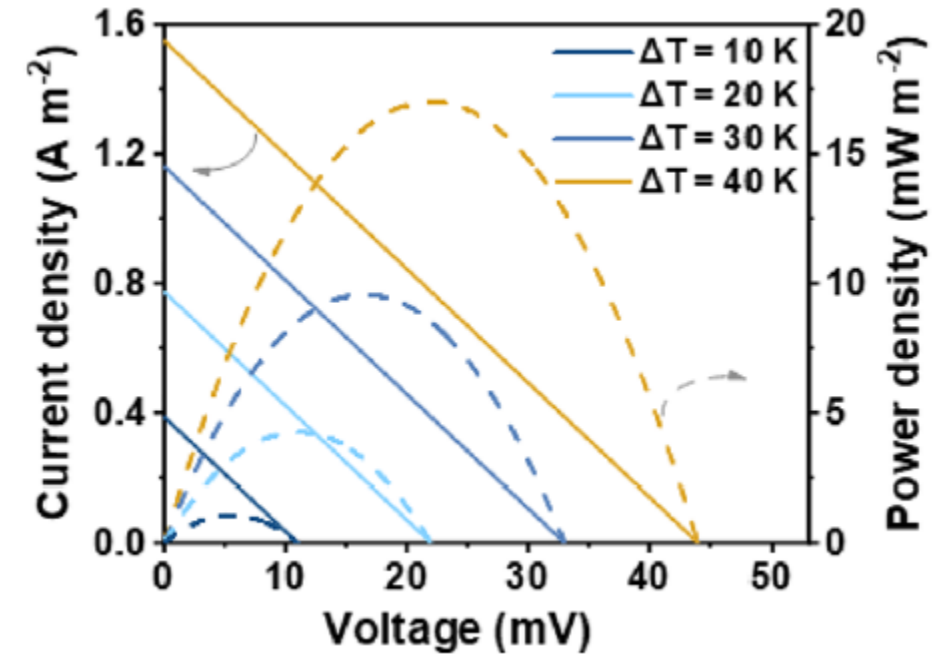
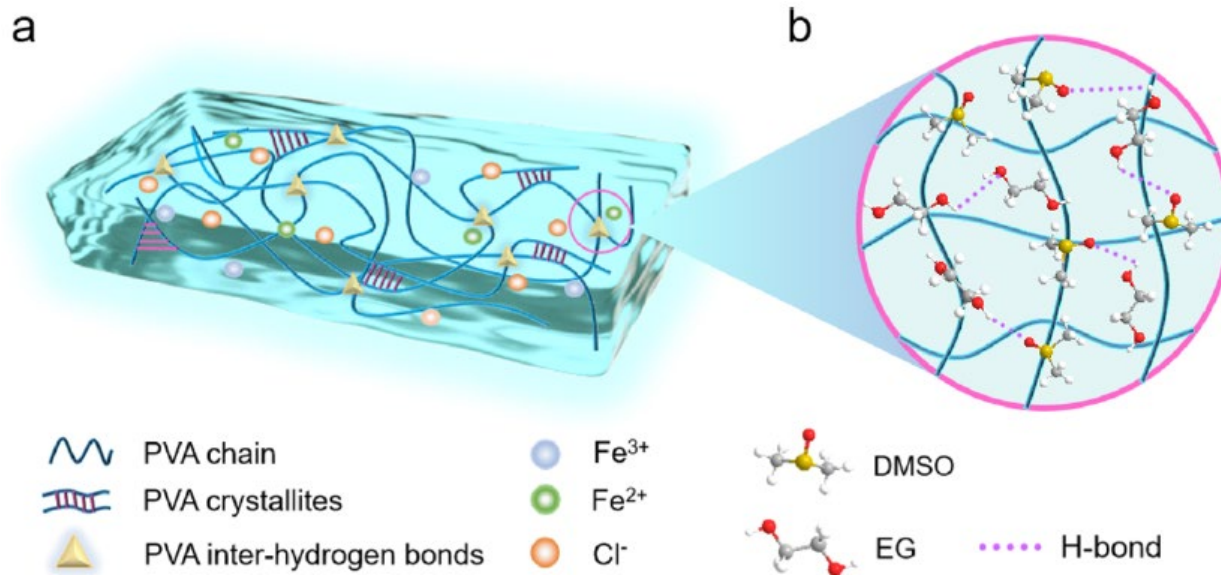
Poor thermal
interfaces



Reclamation potential limited to certain heat flow paths

Thermogalvanic gels readily integrable

Very low power density



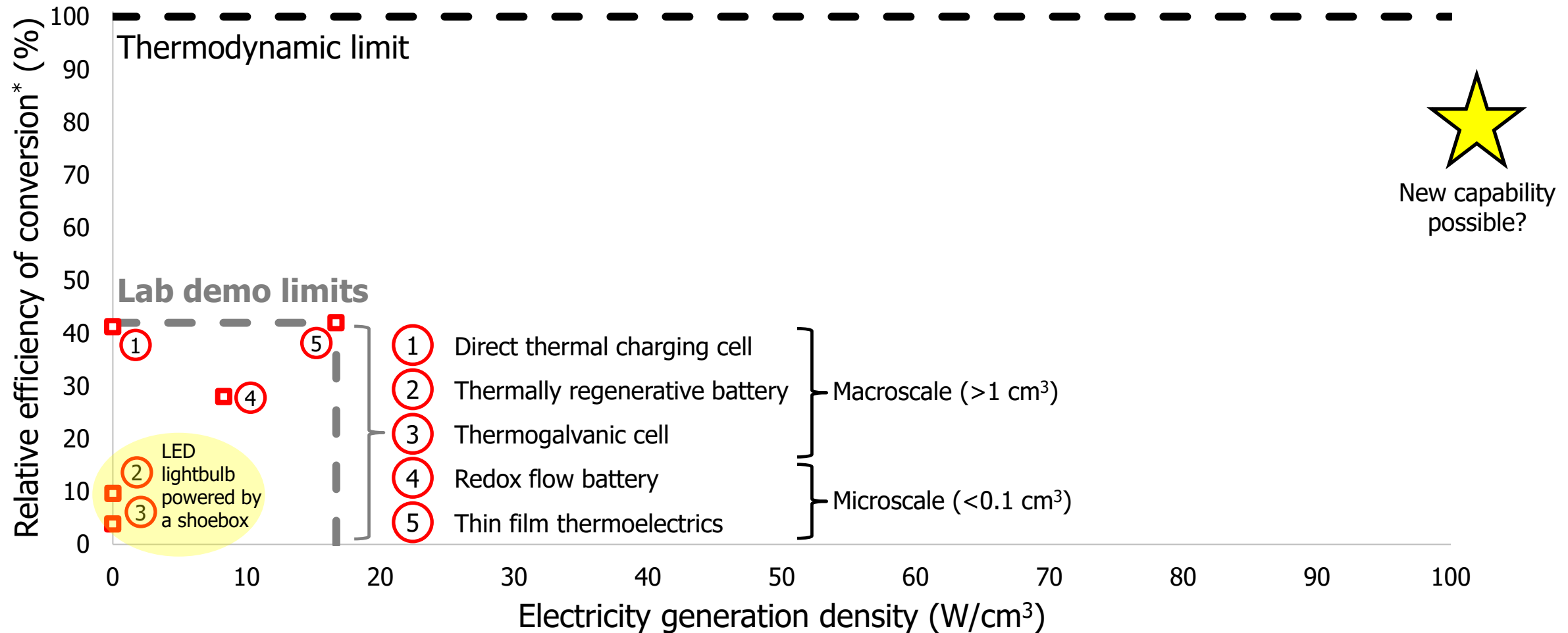


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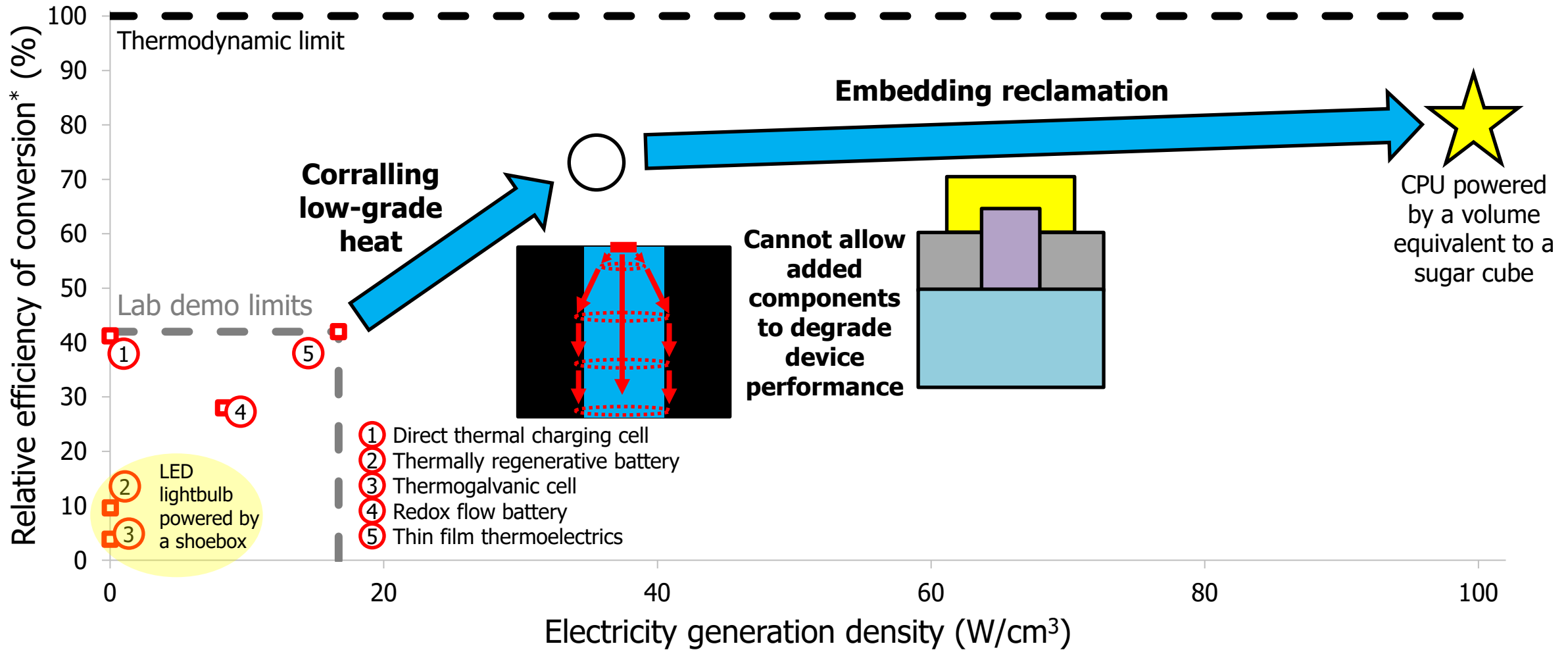
Lab-demonstrated low-grade reclamation efficiency is terrible



None of the low-grade heat is reclaimed because it's seen as not worth it... for now



Electricity generation from low grade waste heat



CPU: Central processing unit

*Relative efficiency is the ratio of the conversion efficiency to the ideal efficiency at max power

M. Wang et al., *Thermal Science and Engineering Progress*, 2023

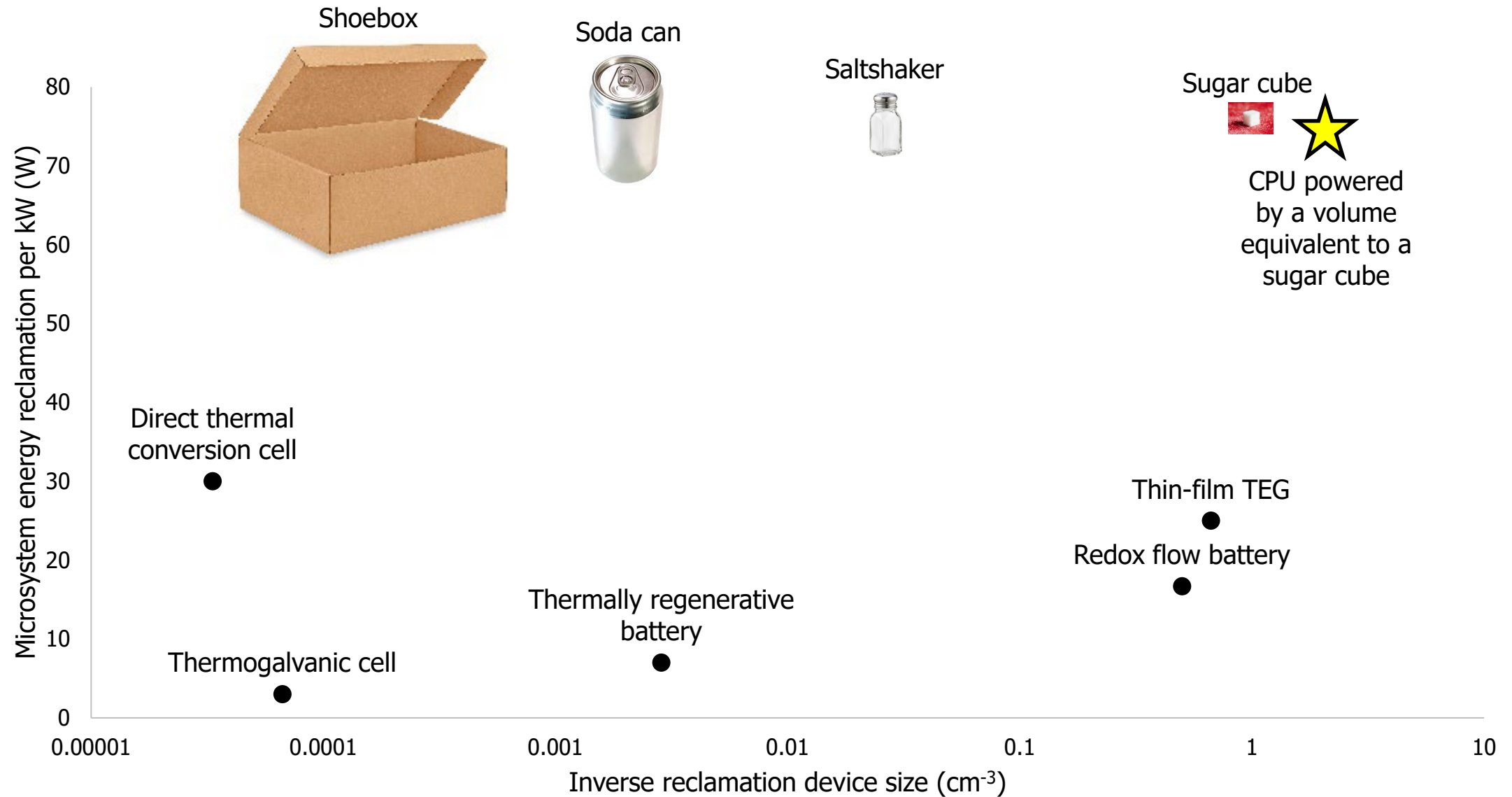
X. Wang et al., *Nature Communications*, 2019

Y. Zhang et al., *Chinese Chemical Letters*, 2023

Distribution Statement A – Approved for Public Release, Distribution Unlimited



Microsystem heat recovery requires recovery in a small form





What parameters could be used to evaluate reclamation performance?

Purpose	Parameter	State-of-the-art	What's possible?
Corralling low-grade heat	Relative efficiency of conversion	~40%	80-90% or higher?
	Thermal resistance ($^{\circ}\text{C}\text{-cm}^2/\text{W}$)	16	0.2 or lower?
	Dynamic responsiveness (Hz)	1×10^2	1×10^4 or greater?
Embedding reclamation	Electrical generation density (W/cm^3)	~10	100 or greater?
Device functionality	Performance degradation	Never tried	Zero impact?

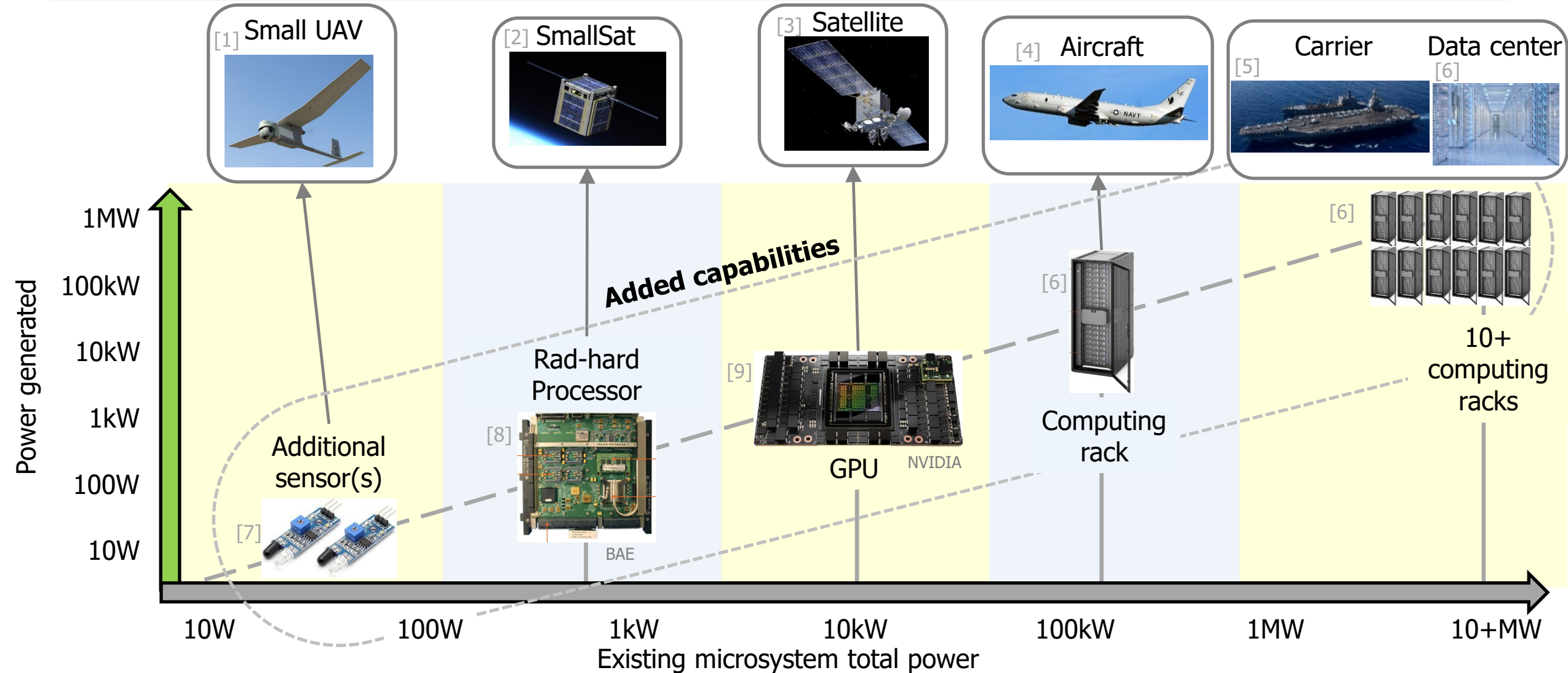


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Energy reclamation enables new capabilities anywhere microsystems can be found



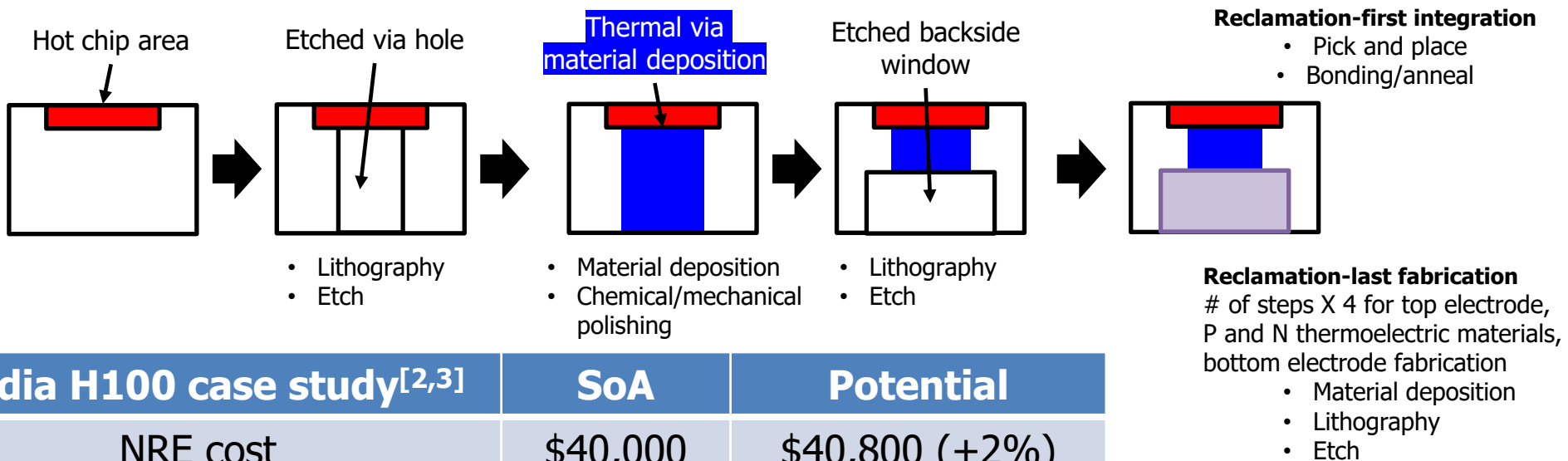
If low-grade heat has all this potential, why don't we reclaim it today?



Economic Analysis

Additional integration steps required (estimate):

- ~4-10 back-end-of-line fabrication steps
- ~0.25% additional fabrication steps of a 4000-step FinFET^[1] technology
 - **< 2% overall chip fabrication cost (upper estimate)**



Nvidia H100 case study ^[2,3]	SoA	Potential
NRE cost	\$40,000	\$40,800 (+2%)
Annual electricity cost ^[a]	\$500	\$500
Cost per compute unit	\$40,500	\$28,300 (-30%) ^[b]
Scaled up to a data center ^[c]	\$405M	\$283M

[a] Paying US average commercial power rate of \$0.127/kWh [4]

[b] 1.5x improvement in computational efficiency

[c] ~5MW compute data center

[1] Intel, 2021

[2] Sarah Jackson, *Business Insider*, 2024

[3] Harry Fowle, *Electronic Specifier*, 2024

[4] US EIA, 2024



www.darpa.mil